

Manual Change History (3666239) Signature™, ISX, and QSX15 Service Manual

Date	16-Aug-2010
Section	Section 6 - Injector and Fuel Lines - Group 06
Procedure Number	10-006-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
Procedure Title	Injector
Update Reason	Update to aftertreatment injector leak test.
Date	17-Sep-2010
Section	Section 6 - Injector and Fuel Lines - Group 06
Procedure Number	10-006-015-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-015-tr.html)
Procedure Title	Fuel Filter (Spin-On Type)
Update Reason	Added content for not prefilling fuel filter.
Date	28-Jun-2010
Section	Section 11 - Exhaust System - Group 11
Procedure Number	101-011-054-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-054-tr.html)
Procedure Title	Aftertreatment Fuel Injector Flow
Update Reason	Initial Check section revised.
Date	11-Oct-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t027-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t027-tr.html)
Procedure Title	Crankcase Gases (Blowby) Excessive
Update Reason	Revision to troubleshooting node - Turbocharger Oil Seal is Leaking.

Date	29-Oct-2010
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-022-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html)
Procedure Title	EGR Valve
Update Reason	Updated instructions for cleaning EGR valve due to active Fault Code 1896.
Date	19-Nov-2010
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-020 (/qs3/pubsys2/xml/en/procedures/101/101-012-020.html)
Procedure Title	Air Compressor (Oil Carryover)
Update Reason	Released this new common procedure.
Date	13-Dec-2010
Section	Section V - Specifications
Procedure Number	10-018-018-tr (/qs3/pubsys2/xml/en/procedures/10/10-018-018-tr.html)
Procedure Title	Cooling System
Update Reason	Updated procedure.
Date	10-Dec-2010
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Completed major edits to multiple steps and sections, including multiple graphic revisions.

Date	11-Jan-2011
Section	Section 16 - Mounting Adaptations
Procedure Number	10-016-005-tr (/qs3/pubsys2/xml/en/procedures/10/10-016-005-tr.html)
Procedure Title	Flywheel
Update Reason	Updated procedure with new part number for service tool.

Date	24-Jan-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t021-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t021-tr.html)
Procedure Title	Coolant Loss - Internal
Update Reason	Updated procedure references and effectivity.

Date	24-Jan-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t025-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t025-tr.html)
Procedure Title	Coolant in the Lubricating Oil
Update Reason	Updated procedure references and effectivity.

Date	28-Jan-2011
Section	Sectopm TT - Troubleshooting Symptoms
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX with CM871
Update Reason	Link fixed for Procedure 101-011-049, added tr effectivity (2C and 10D).

Date	28-Jan-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t092 (/qs3/pubsys2/xml/en/manual/3666239/3666239-t092.html)
Procedure Title	Fuel in the Lubricating Oil
Update Reason	Revised cause and correction 3 new cause and correction 6.
Date	10-Feb-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-047-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html)
Procedure Title	Piston Rings
Update Reason	Updated procedure to reflect current single-piece piston ring gap specifications.
Date	21-Feb-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t078 (/qs3/pubsys2/xml/en/manual/3666239/3666239-t078.html)
Procedure Title	Engine Will Not Crank or Cranks Slowly (Electric Starter)
Update Reason	Tree t078 removed and replaced by t074-005 and t074-010.
Date	21-Feb-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t074-005 (/qs3/pubsys2/xml/en/manual/3666239/3666239-t074-005.html)
Procedure Title	Engine Will Not Crank (Electric Starter)
Update Reason	Replaces t078.

Date	21-Feb-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t074-010 (/qs3/pubsys2/xml/en/manual/3666239/3666239-t074-010.html)
Procedure Title	Engine Cranks Slowly (Electric Starter)
Update Reason	Replaces t078.
Date	09-Mar-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Procedure modified to remove on engine pressure test and replace it with visual inspection criteria.
Date	18-Mar-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t081-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t081-tr.html)
Procedure Title	Engine Will Not Shut Off
Update Reason	Added procedure reference for performing the injector leak test.
Date	21-Mar-2011
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	70-t02-1001 (/qs3/pubsys2/xml/en/procedures/70/70-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree - Signature™ and ISX CM870
Update Reason	Revise Step 9D.

Date	21-Mar-2011
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX with CM871
Update Reason	Revise Step 9F.
Date	21-Mar-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t108-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t108-tr.html)
Procedure Title	Lubricating or Transmission Oil in the Coolant
Update Reason	Revised order of steps in tree.
Date	25-Apr-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t092-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t092-tr.html)
Procedure Title	Fuel in the Lubricating Oil
Update Reason	Steps in tree were modified for internal fuel pump leaks.
Date	06-Jun-2011
Section	Section 1 - Cylinder Block 01
Procedure Number	10-001-145 (/qs3/pubsys2/xml/en/procedures/10/10-001-145.html)
Procedure Title	Variable Geometry Turbocharger Shroud Plate
Update Reason	Corrected the part number for the nozzle vane wear gauge.

Date	06-Jun-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-047-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html)
Procedure Title	Piston Rings
Update Reason	Install step, Single-Piece Piston Ring Gap, changed before and after ESN.
Date	06-Jun-2011
Section	Section 8 - Cooling System - Group 08
Procedure Number	10-008-047 (/qs3/pubsys2/xml/en/procedures/10/10-008-047.html)
Procedure Title	Radiator Pressure Cap
Update Reason	Revised graphic of radiator pressure cap.
Date	14-Jun-2011
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX with CM871
Update Reason	Added Step 5F (percent load check) and removed Step 5C-4.
Date	17-Jun-2011
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	10-006-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
Procedure Title	Injector
Update Reason	Added Note to Finishing Steps: Note: If the injuector o-rings are being replaced due to internal coolant leak, the crankcase breather element mus be changed. Refer to Procedure 003-019 in Section 3.

Date	22-Jun-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	The statement that the shim is inserted on the load side of the gear teeth was removed from the last stepblock of the Backlash (Non-Scissor Gear Combinations) part of the Adjust section.
Date	27-Jun-2011
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Revised the Automotive with CM871 Leak Test procedure to pressure test for a coolant leak.
Date	28-Jun-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t102-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Added inspection of valve stem seals to symptom tree.
Date	11-Jul-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	101-011-042-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-042-tr.html)
Procedure Title	Aftertreatment Fuel Injector
Update Reason	Updated procedure to advise replacing bolts instead of using bolts again, and revised torque specification.

Date	11-Jul-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Updated procedure to change the order of operation to install the cam cover plate then the timing wedges.

Date	11-Jul-2011
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	10-006-025 (/qs3/pubsys2/xml/en/procedures/10/10-006-025.html)
Procedure Title	Static Injection Timing
Update Reason	Added this new procedure to the manual.

Date	11-Jul-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t103-56-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t103-56-tr.html)
Procedure Title	Lubricating Oil Filter Plugged
Update Reason	Revised Engine Timing node and moved up in the tree.

Date	11-Jul-2011
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX with CM871
Update Reason	Updated Step 9A to check static timing and refer to new procedure.

Date	11-Jul-2011
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	70-t02-1001 (/qs3/pubsys2/xml/en/procedures/70/70-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree - Signature™ and ISX CM870
Update Reason	Updated Step 9E to check static timing and refer to new procedure.

Date	14-Jul-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-026-tr.html)
Procedure Title	EGR Differential Pressure Sensor Tubes
Update Reason	Updated: Clean and Inspect for Reuse, Automotive With CM871, Check the inside of the tubes for carbon deposits, and saturate the carbon deposits.

Date	19-Jul-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t004-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t004-tr.html)
Procedure Title	Air Compressor Air Pressure Rises Slowly
Update Reason	Updated procedure links to connect to correct procedures.

Date	15-Aug-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t005-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t005-tr.html)
Procedure Title	Air Compressor Cycles Frequently
Update Reason	Updated procedure links to connect to correct procedures.

Date	19-Jul-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t006-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t006-tr.html)
Procedure Title	Air Compressor Noise is Excessive
Update Reason	Updated procedure links to connect to coorrect procedures.

Date	19-Jul-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t007-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t007-tr.html)
Procedure Title	Air Compressor Pumping Excess Lubricating Oil into the Air System
Update Reason	Updated procedure links to connect to coorrect procedures.

Date	19-Jul-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t008-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t008-tr.html)
Procedure Title	Air Compressor Will Not Maintain Adequate Air Pressure (Not Pumping Continuously)
Update Reason	Updated procedure links to connect to coorrect procedures.

Date	19-Jul-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t009-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t009-tr.html)
Procedure Title	Air Compressor Will Not Pump Air
Update Reason	Updated procedure links to connect to coorrect procedures.

Date	19-Jul-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t010-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t010-tr.html)
Procedure Title	Air Compressor Will Not Stop Pumping
Update Reason	Updated procedure links to connect to coorect procedures.

Date	19-Jul-2011
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-003-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-003-tr.html)
Procedure Title	Air Compressor Carbon Buildup
Update Reason	Updated procedure to include cylinder head cleaning service tool.

Date	07-Jul-2011
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-007-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html)
Procedure Title	Air Compressor Cylinder Head, Single Cylinder
Update Reason	Updated procedure.

Date	25-Jul-2011
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-013-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-013-tr.html)
Procedure Title	Air Compressor Unloader and Valve Assembly
Update Reason	Changed to a common procedure.

Date	25-Jul-2011
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-014-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-014-tr.html)
Procedure Title	Air Compressor
Update Reason	Updated procedure.
Date	19-Jul-2011
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-017-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-017-tr.html)
Procedure Title	Air Governor (Air Compressor Will Not Pump)
Update Reason	Changed to a common procedure.
Date	15-Aug-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-022-001_11-tr (/qs3/pubsys2/xml/en/procedures/10/10-022-001_11-tr.html)
Procedure Title	Service Tools
Update Reason	Updated Tool Number 2892437 and revised tool art for same.
Date	23-Aug-2011
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	10-006-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
Procedure Title	Injector
Update Reason	In the Finishing Steps: Removed inspection of aftertreatment statement referencing Procedure 014-013 and replaced with a note referencing Bulletin 4021600. Removed changing the crankcase breather element note. Added recovery of aftertreatment system note.

Date	23-Aug-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	101-011-054-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-054-tr.html)
Procedure Title	Aftreatment Fuel Injector Flow
Update Reason	Updated the Flow Test section: Updated the amount of fuel to collect and information regarding the use of a stiff brass brush.

Date	27-Aug-2011
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	10-006-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
Procedure Title	Injector
Update Reason	Readded note removed on 23-Aug regarding changing the crankcase breather element (per Sofia).

Date	30-Aug-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Updated procedure with inspection of aftertreatment injector after a EGR cooler was non-reusable.

Date	15-Sep-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t149-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t149-tr.html)
Procedure Title	Stationary Regeneration - Will Not Complete
Update Reason	Updated tree to monitor closing of EGR valve and inspection of EGR differential pressure sensor.

Date	14-Sep-2011
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated leak test procedure to test using pressure decay instead of inspection for bubbles.

Date	14-Sep-2011
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-031-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
Procedure Title	Cylinder Head
Update Reason	Added note to make sure the oil hole in the pressure cap lines up correctly with the regulator valve.

Date	03-Oct-2011
Section	Section 02 - Cylinder Head - Group 02
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Added recommendation to use the ISX15 cylinder head torque template when installing the ISX cylinder head screws to install step.

Date	05-Oct-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t097-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t097-tr.html)
Procedure Title	Intake Manifold Pressure (Boost) is Below Normal
Update Reason	Updated the 6th node of the tree to reflect more engine models.

Date	07-Oct-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-021-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-021-tr.html)
Procedure Title	EGR Mixer
Update Reason	Updated procedure with information related to the EGR hoses from replacing to inspecting the hoses.

Date	07-Oct-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-025-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-025-tr.html)
Procedure Title	EGR Connection Tubes
Update Reason	Updated procedure with information related to the EGR hoses from replacing to inspecting the hoses.

Date	07-Oct-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Updated graphics in pressure test section for EGR cooler and updated part number of EGR cooler leak test kit from 2882437 to 2892437.

Date	17-Oct-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-056 (/qs3/pubsys2/xml/en/procedures/10/10-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Updated procedure to include instructions on how to check the EGR valve, turbocharger, and fueling actuators which can lead to incomplete regeneration.

Date	18-Oct-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t149-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t149-tr.html)
Procedure Title	Stationary Regeneration - Will Not Complete
Update Reason	Updated nodes 6, 7, and 8 to reflect changes in EGR cooler procedure.

Date	18-Oct-2011
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX with CM871
Update Reason	Added Guided Step 1D, Check engine pressure sensors for accuracy.

Date	20-Oct-2011
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-003-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-003-tr.html)
Procedure Title	Lubricating Oil Cooler
Update Reason	Updated a caution statement in the Clean and Inspect for Reuse section with information related to finding metallic debris and dirt in the oil passages of the cooler.

Date	25-Oct-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t108-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t108-tr.html)
Procedure Title	Lubricating or Transmission Oil in the Coolant
Update Reason	Revised turbocharger bearing housing is cracked or porous step.

Date	26-Oct-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Commonize procedure notes and direction surrounding the EGR cooler and EGR differential pressure sensor tubes.

Date	26-Oct-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-026-tr.html)
Procedure Title	EGR Differential Pressure Sensor Tubes
Update Reason	Commonize procedure notes and direction surrounding the EGR cooler and EGR differential pressure sensor tubes.

Date	05-Dec-2011
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t106-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t106-tr.html)
Procedure Title	Lubricating Oil Sludge in the Crankcase Excessive
Update Reason	Updated last node to point to Static Timing procedure 006-025.

Date	05-Dec-2011
Section	Section 11 - Exhaust System - Group 11
Procedure Number	101-011-049-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)
Procedure Title	Aftertreatment Diesel Oxidation Catalyst
Update Reason	Clean and Inspect Reuse Step, added statement, If the aftertreatment diesel oxidation catalyst has a side inlet configuration, insert a shop light and an inspection mirror into the inlet of the diesel oxidation catalyst to aid inspection.

Date	14-Dec-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-043 (/qs3/pubsys2/xml/en/procedures/10/10-001-043.html)
Procedure Title	Piston
Update Reason	Added a note to the procedure to cover the two different single piece piston designs and their compatibility with each other.

Date	18-Jan-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-017-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-017-tr.html)
Procedure Title	Crankshaft Adapter
Update Reason	Install Step, added the word adapter to: To check the installation, attempt to insert a 0.25-mm [0.010-in] feeler gauge between the crankshaft adapter and the tone wheel at four locations approximately 90 degrees apart.

Date	26-Jan-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t102-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Step added to check the static timing.

Date	02-Feb-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Updated procedure to separate base engine timing check.

Date	02-Feb-2012
Section	Section 06 - Injectors and Fuel Lines
Procedure Number	10-006-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
Procedure Title	Injector
Update Reason	Revised Leak Test, If no bubbles are observed in the container while barring the engine it does not indicate that there is no leak. Continue to bar the engine over to build sufficient back pressure to determine which injector is leaking.
Date	10-Feb-2012
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Added Step Intial Check to check liner protrusion and revised Measure Step to check liner protrusion after install.
Date	10-Feb-2012
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Updated manual with new cylinder block counterbore information.
Date	22-Feb-2012
Section	Section 02 - Cylinder Head - Group 02
Procedure Number	10-002-024-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-024-tr.html)
Procedure Title	Overhead Camshaft, Valve
Update Reason	Added a camshaft compatibility matrix and cylinder head cleaning to the Clean and Inspect for Reuse step.

Date	22-Feb-2012
Section	Section 02 - Cylinder Head - Group 02
Procedure Number	10-002-025-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-025-tr.html)
Procedure Title	Overhead Camshaft, Injector
Update Reason	Added a camshaft compatibility matrix and cylinder head cleaning to the Clean and Inspect for Reuse step.

Date	22-Feb-2012
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	10-003-009-tr (/qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html)
Procedure Title	Rocker Lever Assembly
Update Reason	Added a camshaft compatibility matrix and cylinder head cleaning to the Clean and Inspect for Reuse step.

Date	21-Feb-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t027-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t027-tr.html)
Procedure Title	Crankcase Gases (Blowby) Excessive
Update Reason	Removed "Rocker lever housing gasket is leaking" step. Rocker lever housing is not used on ISX, QSX engines.

Date	01-Mar-2012
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	10-003-001-tr (/qs3/pubsys2/xml/en/procedures/10/10-003-001-tr.html)
Procedure Title	Crankcase Breather, External
Update Reason	Added a stepblock to Remove Step to unplug and plug the heater wire harness on an engine that is equipped with a breather heater.

Date	07-Mar-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Revised Inspect for Reuse: If liner ledge pitting, fretting, or wear is discovered block inspection, the block must be machined.

Date	14-Mar-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t149-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t149-tr.html)
Procedure Title	Stationary Regeneration - Will Not Complete
Update Reason	Updated, added technical content, and reorganized steps.

Date	20-Mar-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Updated procedure to clarify measurement requirements.

Date	20-Mar-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Updated procedure to clarify measurement requirements.

Date	26-Mar-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t092-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t092-tr.html)
Procedure Title	Fuel in the Lubricating Oil
Update Reason	Revised Guided Steps in Injector(s) are malfunctioning, Correction

Date	03-Apr-2012
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-056 (/qs3/pubsys2/xml/en/procedures/10/10-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Revise DEF specification and add diagnostic information regarding high resistance in the aftertreatment sensor harness signal wires.

Date	19-Apr-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Updated procedure with a note for run-in piston rings.

Date	19-Apr-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-043 (/qs3/pubsys2/xml/en/procedures/10/10-001-043.html)
Procedure Title	Piston
Update Reason	Updated procedure with a note for run-in piston rings.

Date	19-Apr-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-047-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html)
Procedure Title	Piston Rings
Update Reason	Updated procedure with a note for run-in piston rings.

Date	19-Apr-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-054-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Updated procedure with a note for run-in piston rings.

Date	26-Mar-2012
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Updated procedure to match warranty progressive damage limits.

Date	10-Dec-2009
Section	Section L - Service Literature
Procedure Number	10-205-001-tr (/qs3/pubsys2/xml/en/procedures/10/10-205-001-tr.html)
Procedure Title	Additional Service Literature
Update Reason	

Date	01-May-2012
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	New torque value is added for a new turbocharger stud being announced in a TSB.
Date	01-May-12
Section	Section 14 - Engine Testing - Group 14
Procedure Number	101-014-013-tr (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)
Procedure Title	Aftertreatment Testing
Update Reason	Updated procedure by removing information from Preparatory Steps, moving a test step and adding a test step.
Date	31-May-2012
Section	Section 8 - Cooling System
Procedure Number	10-008-013-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-013-tr.html)
Procedure Title	Coolant Thermostat
Update Reason	Leak Test Step: Added the word "Some" to first sentence. Added (s) to thermostat(s) and seal(s) and changed graphics.
Date	06-Jun-2011
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-007 (/qs3/pubsys2/xml/en/procedures/10/10-007-007.html)
Procedure Title	Lubricating Oil Cooler Element
Update Reason	Added information to the Assemble Step for the new oil cooler element that was recently released.

Date	30-May-2012
Section	Section DS - Engine Disassembly Group 00
Procedure Number	10-001-028-shopds (/qs3/pubsys2/xml/en/procedures/10/10-001-028-shopds.html)
Procedure Title	Cylinder Liner
Update Reason	
Date	30-May-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	A graphic change was made in the Flat Top Piston section. The magnified image was changed to show a flat top cylinder liner image. A word change was made in this section from recession to protrusion.
Date	13-Mar-2002
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-057 (/qs3/pubsys2/xml/en/procedures/10/10-010-057.html)
Procedure Title	Intake Manifold Pressure
Update Reason	
Date	15-Jun-2012
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-057 (/qs3/pubsys2/xml/en/procedures/10/10-010-057.html)
Procedure Title	Turbocharger Wastegate Actuator
Update Reason	Updated the instructions and specifications to measure the actuator rod travel.

Date	12-Jun-2012
Section	Section TT - Troubleshooting Symptoms
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX CM871
Update Reason	Guided Step 2C, Conditions and Action, was changed.

Date	22-Jun-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-014-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-014-tr.html)
Procedure Title	Connecting Rod
Update Reason	Updated procedure by adding new rod bend and twist specifications.

Date	01-Jun-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t170-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t170-tr.html)
Procedure Title	Engine Noise Excessive - Drive Belt
Update Reason	Tree t170 is added.

Date	03-Jul-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t102-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Added 4 new nodes: Lubricating oil cooler is leaking, Oil carryover is excessive because oil pan is regularly overfilled or Lubricating oil dipstick calibration is not correct, Blowby is excessive, and piston, cylinder liner, or piston rings are worn or damaged.

Date	16-Jul-2012
Section	Section 8 - Cooling System - Group 08
Procedure Number	10-008-018-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
Procedure Title	Cooling System
Update Reason	Added new step (Coolant Replacer Tool Method) to perform cooling system service using the Coolant Replacer Tool, Part Number 2892459.
Date	31-Jul-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-043-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-043-tr.html)
Procedure Title	Piston
Update Reason	Added a stepblock In Clean and Inspect For Reuse section to inspect the piston cooling gallery plates.
Date	28-Jul-2012
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	70-t02-1001 (/qs3/pubsys2/xml/en/procedures/70/70-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree - Signature™ and ISX CM870
Update Reason	Changed wastegate controller resistance test values to 6 to 10 ohms, 12-VDC and 24 to 40 ohms for 24-VDC.

Date	02-Aug-2012
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	10-006-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
Procedure Title	Injectors
Update Reason	Leak Test Step: Added, If no bubbles are observed in the container, there is not a leak in the front bank. Continue on to the checks for the rear three cylinders outlined in the procedure below. If bubbles are observed in the container, proceed with barring over the engine to determine which injector is leaking.
Date	15-Aug-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t021-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t021-tr.html)
Procedure Title	Coolant Loss - Internal
Update Reason	Revised "Air compressor cylinder head is cracked or porous, or has a leaking gasket" node
Date	22-Aug-2012
Section	Section 2 - Cylinder Head
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Note remove from Finishing Steps about if the cylinder head or cylinder head gasket is being replaced.

Date	31-Aug-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Added General Information section about hardened brass production liner shim and updated Remove, Clean and Inspect, and Install sections.
Date	04-Sept-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Updated procedure to clarify counterbore machining requirements.
Date	06-Sep-2012
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Updated procedure to include progressive damage inspection of air compressor air inlet and outlet tubes for coolant; if found, inspect air compressor head.
Date	06-Sep-2012
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-007 (/qs3/pubsys2/xml/en/procedures/10/10-007-007.html)
Procedure Title	Lubricating Oil Cooler Element
Update Reason	Updated procedure to distinguish between torque specs for stainless and aluminum

Date	06-Sep-2012
Section	Section DS - Engine Disassembly Group 00
Procedure Number	10-007-003-shopds (/qs3/pubsys2/xml/en/procedures/10/10-007-003-shopds.html)
Procedure Title	Lubricating Oil Cooler
Update Reason	Procedure is updated to include ISX CM871 engine information.
Date	26-Sep-2012
Section	Section 11 - Exhaust System
Procedure Number	10-011-022-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-022-tr.html)
Procedure Title	EGR Valve
Update Reason	Procedure updated to include connecting the EGR valve after cleaning and check for Fault Code 1896.
Date	03-Oct-2012
Section	Section 11 - Exhaust System
Procedure Number	101-011-041-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-041-tr.html)
Procedure Title	Aftertreatment Diesel Particulate Filter
Update Reason	Procedure updated to include statement regarding the proper match of ECM calibrations and DPF part numbers.
Date	10-Oct-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Updated procedure to include caution statements and notes regarding factory shimmed cylinder liners.

Date	24-Oct-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Added to Machine Step: One-half drive ratchet and 15/16 inch socket can be used in place of T-handle for counterbore drive unit rotation.

Date	30-Oct-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-005-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-005-tr.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	Added general information Step and in Install Step added stepblocks to explain saw cut and fracture split connecting rod.

Date	30-Oct-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-014-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-014-tr.html)
Procedure Title	Connecting Rod
Update Reason	Added general information Step and in Install Step added stepblocks to explain saw cut and fracture split connecting rod.

Date	30-Oct-2012
Section	Section 1 - Cylinder Block
Procedure Number	10-001-054-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Added general information Step and in Install Step added stepblocks to explain saw cut and fracture split connecting rod.

Date	20-Nov-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t091-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t091-tr.html)
Procedure Title	Fuel in Coolant
Update Reason	Removed node: Broken valve seat insert. Added node: Bulk coolant supply is contaminated. Revised nodes: Cylinder head is cracked or porous and Injector o-rings or sleeves are damaged. Injector o-rings are missing. New node: Air or combustion gases are entering the cooling system.
Date	30-Nov-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t006-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t006-tr.html)
Procedure Title	Air Compressor Noise is Excessive
Update Reason	Added: Air compressor is excessively worn or internally damaged.
Date	05-Dec-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	A note is added in the Inspect for Reuse step to include the recommendation that the purchase of idler gear with bushing is recommended if being replaced because bushings purchased separately require machining after the installation of the bushing into the gear.

Date	12-Dec-2012
Section	Section 2 - Cylinder Head
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Added sentences to inspect the charge-air cooler if cracks are found in certain areas of the cylinder head in the General Information, Clean and Inspect for Reuse, and Magnetic Crack Inspection steps

Date	11-Jan-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t104 (/qs3/pubsys2/xml/en/manual/3666239/3666239-t104.html)
Procedure Title	Lubricating Oil Pressure High
Update Reason	Changed Cause to: Lubricating oil high-pressure relief valve, oil pressure regulator, or lubricating oil pump os malfunctioning, Correction to: Remove and inspect the high-pressure relief valve, oil pressure regulator, or lubricating oil pump.

Date	25-Jan-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-081-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-081-tr.html)
Procedure Title	Bracket, Retaining Brace
Update Reason	Updated the torque values to match the new engine build standard. Previous value was too high for the aluminum gear housing and too low for the cylinder head.

Date	25-Jan-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Added cylinder liner roundness specifications to procedure.

Date	05-Feb-2013
Section	TF - Troubleshooting Symptoms (New Format)
Procedure Number	105-t02-1005 (/qs3/pubsys2/xml/en/procedures/105/105-t02-1005.html)
Procedure Title	Vibration Excessive
Update Reason	Revised Guided Step 1E, YES, to: Refer to the Engine Runs Rough troubleshooting symptom tree in Section TS or the Engine Performance Troubleshooting Tree in Section TT.

Date	13-Feb-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-018 (/qs3/pubsys2/xml/en/procedures/10/10-001-018.html)
Procedure Title	Crankshaft Gear, Front (Crankshaft Installed)
Update Reason	Updated with instructions on how to clean the old Loctite™ off the crankshaft and instructions on applying Loctite™ when performing a repair where the crankshaft gear is removed.

Date	28-Feb-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t004-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t004-tr.html)
Procedure Title	Air Compressor Air Pressure Rises Slowly
Update Reason	Improved and identified troubleshooting steps implemented in air complaint system trees.

Date	28-Feb-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t005-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t005-tr.html)
Procedure Title	Air Compressor Cycles Frequently
Update Reason	Improved and identified troubleshooting steps implimented in air complaint system trees.

Date	28-Feb-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t006-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t006-tr.html)
Procedure Title	Air Compressor Noise is Excessive
Update Reason	Improved and identified troubleshooting steps implimented in air complaint system trees.

Date	28-Feb-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t008-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t008-tr.html)
Procedure Title	Air Compressor Will Not Maintain Adequate Air Pressure (Not Pumping Continuously)
Update Reason	Improved and identified troubleshooting steps implimented in air complaint system trees.

Date	28-Feb-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t009-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t009-tr.html)
Procedure Title	Air Compressor Will Not Pump Air
Update Reason	Improved and identified troubleshooting steps implimented in air complaint system trees.

Date	28-Feb-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t010-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t010-tr.html)
Procedure Title	Air Compressor Will Not Stop Pumping
Update Reason	Improved and identified troubleshooting steps implimented in air complaint system trees.

Date	28-Feb-2013
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-007-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html)
Procedure Title	Air Compressor Cylinder Head, Single Cylinder
Update Reason	Improved direction on cleaning the air compressor cylinder head.

Date	28-Feb-2013
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-013-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-013-tr.html)
Procedure Title	Air Compressor Unloader and Valve Assembly
Update Reason	Improved direction on cleaning the air compressor cylinder head.

Date	28-Feb-2013
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-017-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-017-tr.html)
Procedure Title	Air Governor (Air Compressor Will Not Pump)
Update Reason	Improved direction on cleaning the air compressor cylinder head.

Date	27-Mar-2013
Section	Section 5 - Fuel System - Group 05
Procedure Number	10-005-239-tr (/qs3/pubsys2/xml/en/procedures/10/10-005-239-tr.html)
Procedure Title	Timing Actuator Flow Test
Update Reason	Added new procedure to manual.
Date	27-Mar-2013
Section	Sectopm TT - Troubleshooting Symptoms
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX with CM871
Update Reason	Added "Perform the Timing Actuator Flow Test" to Guided Step 5.
Date	01-Apr-2013
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t091-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t091-tr.html)
Procedure Title	Fuel in Coolant
Update Reason	Removed: Air or combustion gases are entering the cooling system.
Date	18-Apr-2013
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-145 (/qs3/pubsys2/xml/en/procedures/10/10-010-145.html)
Procedure Title	Variable Geometry Turbocharger Shroud Plate
Update Reason	Corrected part number to 2892212 for CM870 nozzle vane wear gauge.

Date	27-Mar-2013
Section	Sectopm TT - Troubleshooting Symptoms
Procedure Number	16-t02-1001a (/qs3/pubsys2/xml/en/procedures/16/16-t02-1001a.html)
Procedure Title	Engine Performance Troubleshooting Tree - ISX with CM871
Update Reason	Removed from Step 6E a malfunction cause statement naming debris exiting the engine as the source.

Date	01-May-2013
Section	Section 10 - Air Intake System
Procedure Number	10-010-041-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-041-tr.html)
Procedure Title	Turbocharger Coolant Hoses
Update Reason	Remove statements in Install Step: Install the four new banjo sealing washers on the turbocharger coolant supply hose, and Install four new banjo sealing washers on the turbocharger bearing housing coolant return hose and turbocharger actuator coolant return tube.

Date	05-May-2013
Section	Section 1 - Cylinder Block
Procedure Number	10-001-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	A statement is added in the Inspect step to instruct users that all 6 cylinders must be machined if fretting or wear is present on any cylinder.

Date	05-Jun-2013
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-027 (/qs3/pubsys2/xml/en/procedures/10/10-001-027.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	A statement is added in the Inspect step to instruct users that all 6 cylinders must be machined if fretting or wear is present on any cylinder.

Date	05-Jun-2013
Section	Section 1 - Cylinder Liner
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	A statement is added in the Inspect step to instruct users that all 6 cylinders must be machined if fretting or wear is present on any cylinder.

Date	12-Jun-2013
Section	Section 1 - Cylinder Liner
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Added to Inspect for Reuse: Inspect the thrust bearing(s) for pitting, cracking, scratching, uneven wear, or other damage. Replace the thrust bearing(s) and inspect the corresponding gear if damage is found.

Date	25-Jun-2013
Section	Section 2 - Cylinder Head
Procedure Number	10-002-016 (/qs3/pubsys2/xml/en/procedures/10/10-002-016.html)
Procedure Title	Valve Guide Seal, Cylinder Head
Update Reason	Add to General Information: Engines built after ESN 79560433 use a different valve guide seal that requires the use of installation tool Part Number 5296545.
Date	25-Jun-2013
Section	Section 2 - Cylinder Head
Procedure Number	10-22-001_02 (/qs3/pubsys2/xml/en/procedures/10/10-22-001_02.html)
Procedure Title	Service Tools
Update Reason	Added Valve Seal Installation Tool Number 5296545.
Date	28-Jun-2013
Section	Section 7 - Lubricating Oil System
Procedure Number	10-007-007 (/qs3/pubsys2/xml/en/procedures/10/10-007-007.html)
Procedure Title	Lubricating Oil Cooler Element
Update Reason	Clean and Inspect for Reuse Step, added Caution in reference to oil passages contaminated with debris or dirt causes damage.
Date	03-Jul-2013
Section	Section 12 - Compressed Air System
Procedure Number	10-012-014-tr (/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)
Procedure Title	Air Compressor
Update Reason	Added to Remove Step: Pin the engine crankshaft. Refer to Procedure 001-088 in Section 1.

Date	21-Jun-2013
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	70-t02-1001 (/qs3/pubsys2/xml/en/procedures/70/70-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree
Update Reason	Revised Step 6H-5 Yes Block to "Repair the turbocharger assembly.", instead of "Replace the turbocharger assembly."
Date	03-Jul-2013
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	3666239-t087-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	Removed Cause: VE/VMS® analysis uses incorrect data.
Date	14-Aug-2013
Section	Section 11 - Exhaust System
Procedure Number	101-011-047-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-047-tr.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Differential Pressure Sensor Tubes
Update Reason	Revise Install statement: The aftertreatment diesel particulate filter differential pressure sensor is marked with "HI" on one side of the sensor (1). This indicates the high-side differential pressure sensor tube connects to this port. The high-side differential pressure sensor tube connects to the aftertreatment system before the inlet to the diesel particulate filter.

Date	22-Aug-2013
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	10-feisx02 (/qs3/pubsys2/xml/en/procedures/10/10-feisx02.html)
Procedure Title	Fuel Economy Troubleshooting Tree
Update Reason	Removed steps instructing technician to run VE/VMS.

Date	22-Aug-2013
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	10-feisx07 (/qs3/pubsys2/xml/en/procedures/10/10-feisx07.html)
Procedure Title	Fuel Economy Troubleshooting Tree
Update Reason	Removed steps instructing technician to run VE/VMS.

Date	28-Aug-2013
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Update to change the shim part numbers from the obsoleted numbers to the superseded numbers.

Date	02-Oct-2013
Section	Section 5 - Fuel System - Group 05
Procedure Number	10-005-073-tr (/qs3/pubsys2/xml/en/procedures/10/10-005-073-tr.html)
Procedure Title	Integrated Fuel System Module (IFSM)
Update Reason	Updated sensor torque to 14 N.m [124 in-lb].

Date	25-Oct-2013
Section	Section 12 - Compressed Air System
Procedure Number	101-012-008-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-008-tr.html)
Procedure Title	Air Compressor Cylinder Head, Two Cylinder
Update Reason	Added Procedure 012-008 to Section 12.
Date	18-Nov-2013
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	A stepblock and illustration are added to document rotating the turbocharger compressor housing and to include the torque value of the V-band clamp used.
Date	14-Jan-2014
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Added new steps for on-engine EGR cooler testing. Also updated notes about both internal coolant leaks and Fault Codes 555 and/or 556 being present prior to replacing the breather element.
Date	14-Jan-2014
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t021-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t021-tr.html)
Procedure Title	Coolant Loss - Internal
Update Reason	Updated symptom tree to use new EGR cooler troubleshooting and testing procedure.

Date	27-Jan-2014
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Updated Axial and Radial endplay specifications.
Date	27-Feb-2014
Section	Section 0 - Product - Group 00
Procedure Number	10-000-006 (/qs3/pubsys2/xml/en/procedures/10/10-000-006.html)
Procedure Title	Engine Storage - Short Term
Update Reason	Added this procedure to the manual.
Date	17-Mar-2014
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-085 (/qs3/pubsys2/xml/en/procedures/10/10-011-085.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Restriction Test
Update Reason	Added to outline and updated diesel particulate filter restriction test.
Date	05-Mar-2014
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t147-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t147-tr.html)
Procedure Title	Aftertreatment Diesel Particulate Filter - Excessive Automatic and/or Stationary Regeneration
Update Reason	Updated diesel particulate filter restriction test troubleshooting symptom tree with DPF restriction test.

Date	04-Mar-2014
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Updated diesel particulate filter restriction test.

Date	27-Mar-2014
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666230-t103-56-tr (/qs3/pubsys2/xml/en/manual/3666230/3666230-t103-56-tr.html)
Procedure Title	Lubricating Oil Filter Plugged
Update Reason	Add: In-cylinder damage and Engine timing incorrect, to troubleshooting symptom tree.

Date	04-Apr-2014
Section	Section 10 - Air Intake System
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Revised procedure to include a split between CM870 and CM871.

Date	10-Apr-2014
Section	Section 1 - Cylinder Block
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Added information: retainer plate and retainer part no. are both use for the lower idler gear and the adjustable idler gear.

Date	14-Apr-2014
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Updated note in Finishing Steps.
Date	21-Apr-2014
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	10-022-001_06 (/qs3/pubsys2/xml/en/procedures/10/10-022-001_06.html)
Procedure Title	Service Tools
Update Reason	Added 3164197 Service Tool (Injector Seal Replacer Kit).
Date	25-Apr-2014
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	10-003-009-tr (/qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html)
Procedure Title	Rocker Lever Assembly
Update Reason	Corrected error in assembly sequence.
Date	25-Apr-2014
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	101-012-007-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-007-tr.html)
Procedure Title	Air Compressor Cylinder Head, Single Cylinder
Update Reason	This revised head with changes that need to be addressed in the manual to avoid confusion. The revised head has only one spring pin the old head had two.

Date	25-Jun-2014
Section	Section 12 Compressed Air System - Group 12
Procedure Number	101-012-024-tr (/qs3/pubsys2/xml/en/procedures/101/101-012-024-tr.html)
Procedure Title	Air Pressure Relief Valve
Update Reason	Added steps for testing the operation of the air compressor relief valve.
Date	10-Jul-2014
Section	Section 11 - Exhaust System - Group 11
Procedure Number	101-011-042-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-042-tr.html)
Procedure Title	Aftertreatment Fuel Injector
Update Reason	Torque specification update to reflect change in supplier, and change in procedure to dispose of used capscrews.
Date	23-Jul-2014
Section	Section 10 Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Added the clocking procedure.
Date	24-Jul-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-054-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Updated the connecting rod side clearance and updated the Remove Step.

Date	11-Aug-2014
Section	Section 11 - Exhaust System
Procedure Number	10-011-085 (/qs3/pubsys2/xml/en/procedures/10/10-011-085.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Restriction Test
Update Reason	Updated 69 VDC to 0.69 VDC.
Date	28-Aug-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-033-tr.html)
Procedure Title	Gear Housing, Front
Update Reason	Updated procedure for new block plug part number.
Date	28-Aug-2014
Section	Section 16 - Mounting Adaptations
Procedure Number	10-016-006-tr (/qs3/pubsys2/xml/en/procedures/10/10-016-006-tr.html)
Procedure Title	Flywheel Housing
Update Reason	Updated procedure for new block plug part number.
Date	28-Aug-2014
Section	Section 16 - Mounting Adaptations
Procedure Number	10-016-007-tr (/qs3/pubsys2/xml/en/procedures/10/10-016-007-tr.html)
Procedure Title	Flywheel Housing, REPTO
Update Reason	Updated procedure for new block plug part number.

Date	10-Oct-2014
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-024-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-024-tr.html)
Procedure Title	Overhead Camshaft, Valve
Update Reason	Added warnings for handling camshaft when removing, cleaning, and installing camshaft.
Date	10-Oct-2014
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-025-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-025-tr.html)
Procedure Title	Overhead Camshaft, Injector
Update Reason	Added warnings for handling camshaft when removing, cleaning, and installing camshaft.
Date	12-Nov-2014
Section	Section 8 - Cooling System - Group 08
Procedure Number	10-008-013-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-013-tr.html)
Procedure Title	Coolant Thermostat
Update Reason	Statement is added in the Remove step to instruct user to remove both thermostat and seal from the thermostat housing cover and to discard the seal.
Date	07-Nov-2014
Section	Section 6 - Injectors and Fuel Lines
Procedure Number	10-006-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
Procedure Title	Injectors
Update Reason	Reference to the injector clamp capscrew washer that is not needed, is removed.

Date	01-Dec-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Added inspection criteria for machined-from-solid camshaft gear retaining washers.

Date	16-Dec-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Added reference to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753, to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.

Date	16-Dec-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-027 (/qs3/pubsys2/xml/en/procedures/10/10-001-027.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Added reference to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753, to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.

Date	29-Dec-2014
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Added reference to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753, to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.
Date	08-Jan-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Added reference to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753, to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.
Date	08-Jan-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-054-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Added antipolishing ring liner information.
Date	20-Jan-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Added information on antipolishing ring liners for the 15L products.

Date	12-Feb-2015
Section	Section 1 - Cylinder Block
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Corrected location of specification to the counterbore depth in the Machine Section.
Date	16-Feb-2015
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	10-016-007-tr (/qs3/pubsys2/xml/en/procedures/10/10-016-007-tr.html)
Procedure Title	Flywheel Housing, REPTO
Update Reason	Procedure updated to address new idler shaft design and differences in hardware and torque specifications between the two shaft designs.
Date	06-Mar-2015
Section	Section 5 - Fuel System - Group 05
Procedure Number	10-005-016 (/qs3/pubsys2/xml/en/procedures/10/10-005-016.html)
Procedure Title	Fuel Pump
Update Reason	Updated torque and graphic for IFSM fuel gear pump.
Date	16-Mar-2015
Section	Section 1 - Cylinder Block
Procedure Number	10-001-054-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Explained that antipolishing ring service tools are not sold separately and are only available as part of kit Part Number 5299448.

Date	27-Mar-2015
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Created a new turbocharger coolant leak test procedure using the new Turbocharger Coolant Leak Test Kit, Part Number 5299740.
Date	01-Apr-2015
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Added a note and specification that limits the depth of machining to 0.081 mm [0.032 in].
Date	06-Apr-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-014-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-014-tr.html)
Procedure Title	Connecting Rod
Update Reason	Added specification for measuring the capscrews diameter above the threads (checking for necking) and a length specification.
Date	14-Apr-2015
Section	Section 8 - Cooling System - Group 08
Procedure Number	10-008-018-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
Procedure Title	Cooling System
Update Reason	Added direction to let coolant cool before draining.

Date	13-Apr-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-027 (/qs3/pubsys2/xml/en/procedures/10/10-001-027.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Changes made to Leak Test Section of the procedure.
Date	22-Apr-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Changes made to Initial Check (Liner Protrusion) and Measure (Liner Protrusion) Sections of the procedure.
Date	24-Apr-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Removed inspection for discoloration as it is not a reliable indicator for block reuse.
Date	30-Apr-2015
Section	Section 7 - Lubricating Oil System
Procedure Number	10-007-014 (/qs3/pubsys2/xml/en/procedures/10/10-007-014.html)
Procedure Title	Lubricating Oil Filter Bypass Valve
Update Reason	Updated procedure to include single-piece bypass valve.

Date	08-May-2015
Section	Section 1 - Cylinder Block
Procedure Number	10-001-005-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-005-tr.html)
Procedure Title	Bearing, Connecting Rod
Update Reason	Updated Note in Remove step: Both the rod cap and the connecting rod are serialized on one face to make sure of proper orientation.
Date	13-May-2015
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Update to Measurement section to indicate the block must be replaced if the main bearing bore is not within specifications.
Date	19-Jan-2015
Section	Section 1 - Cylinder Block
Procedure Number	10-001-023-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-023-tr.html)
Procedure Title	Crankshaft Seal, Front
Update Reason	Revised to state that the seal sits on the crankshaft adaptor not the crankshaft.
Date	15-May-2015
Section	Section 2 - Cylinder Head
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Revised Remove and Install verbiage related to valve stem and guide seal.

Date	01-Jun-2015
Section	Section 2 - Cylinder Head
Procedure Number	10-002-016 (/qs3/pubsys2/xml/en/procedures/10/10-002-016.html)
Procedure Title	Valve Guide Seal, Cylinder Head
Update Reason	Revised with integrated valve guide seal information.
Date	06-Jul-2015
Section	Section 14 - Engine Testing
Procedure Number	99-014-024 (/qs3/pubsys2/xml/en/procedures/99/99-014-024.html)
Procedure Title	Fluorescent Tracer Dye Test
Update Reason	Added to outline.
Date	13-Jul-2015
Section	Section 20 - Vehicle Braking - Group 20
Procedure Number	10-020-015 (/qs3/pubsys2/xml/en/procedures/10/10-020-015.html)
Procedure Title	Engine Brake Wiring Harness
Update Reason	A mounting tab has been added to the engine brake wiring harness frame. Edited the procedure to include the necessary service instructions.
Date	28-Aug-2015
Section	Section 14 - Engine Testing - Group 14
Procedure Number	101-014-013-tr (/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)
Procedure Title	Aftertreatment Testing
Update Reason	Removed Preparatory Step section. Added regeneration information to Test section.

Date	24-Sep-2015
Section	Section 20 - Vehicle Braking
Procedure Number	10-020-012 (/qs3/pubsys2/xml/en/procedures/10/10-020-012.html)
Procedure Title	Engine Brake Solenoid Valve
Update Reason	Added Inspect for Reuse step.
Date	01-Oct-2015
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	101-003-019-tr (/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)
Procedure Title	Crankcase Breather Element
Update Reason	Added the Clean and Inspect for Reuse step.
Date	12-Nov-2015
Section	Section 0 - Product - Group 00
Procedure Number	10-000-017 (/qs3/pubsys2/xml/en/procedures/10/10-000-017.html)
Procedure Title	Engine Barring
Update Reason	Added to outline.
Date	08-Feb-2016
Section	Section 11 - Exhaust System
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Added a note to indicate mounting capscrews may have washers instead of clamping plates.

Date	25-Feb-2016
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-005-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-005-tr.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	Updated Remove step and Fracture-Split Style Connecting Rod Bearing Thickness specifications.

Date	08-Mar-2016
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Procedure to include recent changes and release activity on the large outside diameter (152mm) liner.

Date	09-Mar-2016
Section	Section 7 - Lubricating Oil System
Procedure Number	10-007-031-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Updated to reflect the removal of the shim information on all current product blocks.

Date	02-May-2016
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-016-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Added a note in the Clean and Inspect for Reuse step.

Date	01-Jun-2016
Section	Section 11 - Exhaust System
Procedure Number	10-011-019-tr (/qs3/pubsys2/xml/en/procedures/10/10-011-019-tr.html)
Procedure Title	EGR Cooler
Update Reason	Added a note concerning the maintenance-free crankcase breather.
Date	15-Jun-2016
Section	Section 8 - Cooling System - Group 08
Procedure Number	10-008-013-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-013-tr.html)
Procedure Title	Coolant Thermostat
Update Reason	Removed a note from Leak Test section regarding use of an electronic service tool for monitoring temperature.
Date	17-Jun-2016
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	10-003-019 (/qs3/pubsys2/xml/en/procedures/10/10-003-019.html)
Procedure Title	Crankcase Breather Element
Update Reason	Added General Information step.
Date	27-Jun-2016
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	10-003-009-tr (/qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html)
Procedure Title	Rocker Lever Assembly
Update Reason	Added information on lubricating roller pins during assembly.

Date	11-Jul-2016
Section	Section 7 - Lubricating Oil System
Procedure Number	10-007-031-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Updated Disassemble and Assemble graphics to clarify pressure regulator parts and orientation. Added note to Assembly steps to specify orientation of pressure regulator end cap.
Date	14-Jul-2016
Section	TF - Troubleshooting Symptoms (New Format)
Procedure Number	105-t02-1005 (/qs3/pubsys2/xml/en/procedures/105/105-t02-1005.html)
Procedure Title	Vibration Excessive
Update Reason	Revised troubleshooting for brake adjustment, engine mounts, front end belt drive accessories, and vibration damper.
Date	08-Aug-2016
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-014-tr-prix13 (/qs3/pubsys2/xml/en/procedures/10/10-007-014-tr-prix13.html)
Procedure Title	Lubricating Oil Filter Bypass Valve
Update Reason	Updated General information step pressure drop value.
Date	26-Aug-2016
Section	Section 5 - Fuel System - Group 05
Procedure Number	10-005-043 (/qs3/pubsys2/xml/en/procedures/10/10-005-043.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	Updated torque value for Allen head capscrew.

Date	08-Sep-2016
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-016 (/qs3/pubsys2/xml/en/procedures/10/10-002-016.html)
Procedure Title	Valve Guide Seal, Cylinder Head
Update Reason	Updated the General Information Section to state that the CM2250 engine takes a specific type of valve guide.

Date	21-Sep-2016
Section	Section 11 - Exhaust System
Procedure Number	10-011-085 (/qs3/pubsys2/xml/en/procedures/10/10-011-085.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Restriction Test
Update Reason	Added DPF part numbers and specifications.

Date	15-Sep-2016
Section	Section 1 - Cylinder Block
Procedure Number	10-001-014 (/qs3/pubsys2/xml/en/procedures/10/10-001-014.html)
Procedure Title	Connection Rod
Update Reason	Added inside diameter specifications for machine and fracture split connecting rods.

Date	20-Jan-2017
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-134 (/qs3/pubsys2/xml/en/procedures/10/10-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Update to Clean and Inspect for Reuse.

Date	07-Mar-2017
Section	Section 2 - Cylinder Head
Procedure Number	10-002-014 (/qs3/pubsys2/xml/en/procedures/10/10-002-014.html)
Procedure Title	Injector Sleeve, Cylinder Head
Update Reason	Added procedure to manual.
Date	28-Mar-2017
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Radial Play measurement for Compressor wheel and Impeller has been replaced to remove incorrect diagnostic.
Date	02-May-2017
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-014 (/qs3/pubsys2/xml/en/procedures/10/10-002-014.html)
Procedure Title	Injector Sleeve, Cylinder Head
Update Reason	Updated Finishing Steps for engine configuration.
Date	16-May-2017
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Update is to remove potential misunderstandings about replacing turbochargers. The first is to Remove "Replace if leak found", the second "cleaning post repair" and the third is CSOL understand as set with precedence in 6.7L CM2350 B101.

Date	26-May-2017
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t105-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t105-tr.html)
Procedure Title	Lubricating Oil Pressure Low
Update Reason	Added node for "Lubrication oil thermostat is malfunctioning".
Date	30-Aug-2017
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated procedure with revised part number for valve vacuum tester.
Date	31-Aug-2017
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-022-001_02 (/qs3/pubsys2/xml/en/procedures/10/10-022-001_02.html)
Procedure Title	Service Tools
Update Reason	Updated procedure with revised part number for valve vacuum tester.
Date	10-Oct-2017
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-014 (/qs3/pubsys2/xml/en/procedures/10/10-002-014.html)
Procedure Title	Injector Sleeve, Cylinder Head
Update Reason	Added Clean and Inspect for Reuse step.

Date	12-Oct-2017
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666239-t091-tr (/qs3/pubsys2/xml/en/manual/3666239/3666239-t091-tr.html)
Procedure Title	Fuel in Coolant
Update Reason	Updated first node cause from "Broke valve seat insert" to "Bulk coolant supply is contaminated".

Date	17-Nov-2017
Section	Section 10 - Air Intake System
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Revised Initial step to include service bulletin and troubleshooting trees.

Date	08-May-2018
Section	Section 1 - Cylinder Block
Procedure Number	10-001-017 (/qs3/pubsys2/xml/en/procedures/10/10-001-017.html)
Procedure Title	Crankshaft Adapter
Update Reason	Updated incorrect crankshaft adapter drill hole specifications in the Install step while updating the procedure to current standards.

Date	18-May-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-013-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-013-tr.html)
Procedure Title	Lubricating Oil Filter (Spin-On)
Update Reason	Updated General Information section for the combination NanoNet® lubricating oil filter.

Date	11-May-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-999 (/qs3/pubsys2/xml/en/procedures/10/10-007-999.html)
Procedure Title	Lubricating Oil System - Overview
Update Reason	Updated Lubricating Oil Filters of the General Information with the combination NanoNet® lubricating oil filter.
Date	14-May-2018
Section	Section V - Specifications
Procedure Number	10-018-017 (/qs3/pubsys2/xml/en/procedures/10/10-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Updated oil filter capacity of standard engine.
Date	01-Jun-2018
Section	Section 2 - Cylinder Head
Procedure Number	10-002-014 (/qs3/pubsys2/xml/en/procedures/10/10-002-014.html)
Procedure Title	Injector Sleeve, Cylinder Head
Update Reason	Updated to direct the field to a head replacement as the warrantable/policy repair practice instead of the newly released Blue o-ring, Part Number 5440257.
Date	06-Jul-2018
Section	Section 7 - Lubricating Oil System
Procedure Number	10-007-083 (/qs3/pubsys2/xml/en/procedures/10/10-007-083.html)
Procedure Title	Lubricating Oil and Filter Analysis
Update Reason	Updated procedure with instruction for inspect and reuse of other engine components if debris is found in the filter.

Date	26-Jul-2018
Section	Section 7 - Lubricating Oil System
Procedure Number	10-007-003-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-003-tr.html)
Procedure Title	Lubricating Oil Cooler
Update Reason	The leak test was changed to a pressure decay test with an acceptable leak rate.
Date	13-Sep-2018
Section	Section 8 - Cooling System
Procedure Number	10-008-102-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-102-tr.html)
Procedure Title	Water Pump Cartridge
Update Reason	Updated capscrew torque specifications.
Date	13-Sep-2018
Section	Section 8 - Cooling System
Procedure Number	10-008-018-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
Procedure Title	Cooling System
Update Reason	Specified rpm operation at high idle.
Date	28-Sep-2018
Section	Section 8 - Cooling System
Procedure Number	10-008-062-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Added automotive CM570 and industrial CM570 with two piece design water pump stepblock.

Date	16-Oct-2018
Section	Section 14 - Engine Testing - Group 14
Procedure Number	10-014-027 (/qs3/pubsys2/xml/en/procedures/10/10-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	This procedure was moved from Section 19 (Procedure 72-019-666) in the Signature™, ISX, and QSX15 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666259, to this manual. All content is the same.
Date	02-Jan-2019
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-031-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Added information covering the pinned and threaded relief valves.
Date	16-Jan-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Updated torque value for lower idler gears.
Date	22-Jan-2019
Section	Section V - Specifications
Procedure Number	10-018-017 (/qs3/pubsys2/xml/en/procedures/10/10-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Updated oil pressure at no-load governed speed.

Date	05-Feb-2019
Section	Section 11 - Exhaust System
Procedure Number	10-011-073 (/qs3/pubsys2/xml/en/procedures/10/10-011-073.html)
Procedure Title	Exhaust Manifold Turbocharger Mounting Stud Replacement
Update Reason	Added to outline.
Date	01-Mar-2019
Section	Section 6 - Injections and Fuel Lines - Group 06
Procedure Number	10-006-012 (/qs3/pubsys2/xml/en/procedures/10/10-006-012.html)
Procedure Title	Fuel Drain Line Restriction
Update Reason	Added a note which explains the applications that this specification would apply to.
Date	11-Mar-2019
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-031-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Modified Install section to differentiate between pumps that use shims to set lash and other pump that does not use shims.
Date	18-Apr-2019
Section	Section 2 - Cylinder Head
Procedure Number	10-002-016 (/qs3/pubsys2/xml/en/procedures/10/10-002-016.html)
Procedure Title	Valve Guide Seal, Cylinder Head
Update Reason	Updated Preparatory/Finishing Steps to remove/install the rocker lever shaft assemblies, and adjust the overhead set as needed.

Date	18-Apr-2019
Section	Section 2 - Cylinder Head
Procedure Number	10-002-025-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-025-tr.html)
Procedure Title	Overhead Camshaft, Injector
Update Reason	Update graphic to show housing/engine of a single camshaft.
Date	18-Apr-2019
Section	Section 3 - Rocker Levers
Procedure Number	10-003-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-003-004-tr.html)
Procedure Title	Overhead Set
Update Reason	Corrected the minimum value for the intake valve and revised graphic in General Information.
Date	22-Apr-2019
Section	Section 3 - Rocker Levers
Procedure Number	10-003-009-tr (/qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html)
Procedure Title	Rocker Lever Assembly
Update Reason	Removed information pertaining to repairing the motor, need to replace.
Date	23-Apr-2019
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-134 (/qs3/pubsys2/xml/en/procedures/10/10-010-134.html)
Procedure Title	Variable Geometry Turbocharger Actuator, Electric
Update Reason	Updated procedure to align the actuator using guide pins from the seravoid potential damage during installation.

Date	06-May-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Added a stepblock at the end of the Inspect for Reuse section to inspect the upper idler gear shaft assembly.

Date	06-May-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-111 (/qs3/pubsys2/xml/en/procedures/10/10-001-111.html)
Procedure Title	Idler Gear, Upper
Update Reason	Added this procedure to the manual.

Date	3-May-2019
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	10-016-007-tr (/qs3/pubsys2/xml/en/procedures/10/10-016-007-tr.html)
Procedure Title	Flywheel Housing, REPTO
Update Reason	Modified rear end power take-off (REPTO) backlash limits.

Date	31-May-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-089-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html)
Procedure Title	Block Stiffener Plate
Update Reason	Updated Torque Values in the Install step.

Date	13-Jun-2019
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-025-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
Procedure Title	Lubricating Oil Pan
Update Reason	Created stepblock specifically for the oil drain plug.

Date	13-Jun-2019
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Added a note to the beginning of the Inspect for Reuse step.

Date	15-Jul-2019
Section	Section 1 - Cylinder Block- Group 01
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Updated procedure to inform the field about brass shims.

Date	10-Sep-2019
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-024-tr- (/qs3/pubsys2/xml/en/procedures/10/10-002-024-tr-.html)
Procedure Title	Overhead Camshaft, Valve
Update Reason	Added Camshaft Removal and Installation Kit, Part Number 5395348, instructions.

Date	16-Aug-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Modify the procedure when resurfacing the mid-stop counterbore.
Date	27-Aug-2019
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-022-001_02 (/qs3/pubsys2/xml/en/procedures/10/10-022-001_02.html)
Procedure Title	Service Tools
Update Reason	Added Camshaft Removal and Installation Kit, Part Number 5395348.
Date	28-Oct-2019
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-025-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-025-tr.html)
Procedure Title	Overhead Camshaft, Injector
Update Reason	Revised graphic for removing the injector camshaft.
Date	17-Dec-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-052-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-052-tr.html)
Procedure Title	Vibration Damper, Viscous
Update Reason	Added stepblock to the Clean and Inspect for Reuse steps to remove the vibration damper from the oven to check for fluid leakage.

Date	15-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-026-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-026-tr.html)
Procedure Title	Cylinder Block
Update Reason	Removed statetment located in the Inspect for Reuse steps that called for all cylinders to be replaced if fretting or wear is present.

Date	15-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-027 (/qs3/pubsys2/xml/en/procedures/10/10-001-027.html)
Procedure Title	Cylinder Block and Liner Seats
Update Reason	Removed statement located in the Leak Test steps that called for all six cylinder liners to be replaced if fretting or wear was present.

Date	24-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-022-001_01 (/qs3/pubsys2/xml/en/procedures/10/10-022-001_01.html)
Procedure Title	Service Tools
Update Reason	Revised the listed part number for the marked socket from 3165161 to 5573072.

Date	24-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-006-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-006-tr.html)
Procedure Title	Bearings, Main
Update Reason	Revised the listed part number for the marked socket from 3165161 to 5573072.

Date	29-Jan-2020
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Added service bulletin resources for damaged turbocharger compressor impellers and turbine wheels.
Date	26-Feb-2020
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-031-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-031-tr.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Corrected the listed specifications for lubricating oil pressure regulator spring load and high-pressure relief valve spring in regards to the regulator spring bore with a threaded regulator plug.
Date	04-Mar-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Revised torque value for the adjustable idler gear listed in the Adjust section of the procedure to the correct values.

Date	16-Mar-2020
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	10-014-027 (/qs3/pubsys2/xml/en/procedures/10/10-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	Updated the procedure to require INSITE service tool version 8.5.2 or later in order to fully support ARA tests.
Date	15-May-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-058-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
Procedure Title	Cylinder Block Counterbore
Update Reason	Added Select Service Tools list to the procedure.
Date	15-May-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-022-001_01 (/qs3/pubsys2/xml/en/procedures/10/10-022-001_01.html)
Procedure Title	Service Tools
Update Reason	Added Counterbore Cutting Improvement Kit, Part Number 5395552 to Select Service Tools list.
Date	18-Jun-2020
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-004-tr (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Made minor corrections to procedure.

Date	14-Aug-2020
Section	Section F - Familiarization
Procedure Number	10-200-001-tr (/qs3/pubsys2/xml/en/procedures/10/10-200-001-tr.html)
Procedure Title	Flow Diagram, Fuel System
Update Reason	Corrected typo found in the procedure.
Date	04-Mar-2021
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	10-002-048 (/qs3/pubsys2/xml/en/procedures/10/10-002-048.html)
Procedure Title	Camshaft Cover Plate
Update Reason	Added procedure to manual.
Date	22-Mar-2021
Section	Section V - Specifications
Procedure Number	10-018-017 (/qs3/pubsys2/xml/en/procedures/10/10-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Corrected listed specification for At No-Load Governed Speed to reflect appropriate values.
Date	20-Jul-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-028-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Corrected maximum specification listed for the Cylinder Liner Protrusion (around individual cylinder).

Date	20-Jul-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-088-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
Procedure Title	Engine Base Timing
Update Reason	Added instruction to rotate the valve camshaft gear mounting capscrew an additional 120 degrees in the Adjust step.

Date	13-Sep-2021
Section	Section 14 - Engine Testing - Group 14
Procedure Number	10-014-027 (/qs3/pubsys2/xml/en/procedures/10/10-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	Revised Select Service Tools list. Removed Analyzing the Data step from the procedure.

Date	23-Nov-2021
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-025-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
Procedure Title	Lubricating Oil Pan
Update Reason	Corrected torque value listed for the oil drain valve.

Date	30-Nov-2021
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	10-007-037-tr (/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
Procedure Title	Lubricating Oil System
Update Reason	Corrected torque value listed for the oil drain plugs.

Date	01-Feb-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-018 (/qs3/pubsys2/xml/en/procedures/10/10-001-018.html)
Procedure Title	Crankshaft Gear, Front (Crankshaft Installed)
Update Reason	Added Crankshaft adapter and gear remover kit, Part Number 5572864, Magnetic base drill, Part Number 5572871 (60 Hz) or 5572872 (50 Hz), and Cutter bit Insert, Part Number 5572875 to Select Service Tools list. Added Remove step that uses magnetic base drill.
Date	01-Feb-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-017-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-017-tr.html)
Procedure Title	Crankshaft Adapter
Update Reason	Added necessary tools to Select Service Tools list. Added Remove steps that utilizes a magnetic base drill.
Date	10-Aug-2022
Section	Section 8 - Cooling System - Group 08
Procedure Number	10-008-036-tr (/qs3/pubsys2/xml/en/procedures/10/10-008-036-tr.html)
Procedure Title	Fan Hub, Belt Driven
Update Reason	Added Disassemble and Assemble steps to procedure.
Date	29-Aug-2022
Section	Section 5 - Fuel System - Group 05
Procedure Number	10-005-043 (/qs3/pubsys2/xml/en/procedures/10/10-005-043.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	Consolidated and improved Fuel Shutoff Valve procedure.

Date	03-Oct-2022
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-033-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Updated turbocharger information for CM871.
Date	05-Dec-2022
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	10-016-005-tr (/qs3/pubsys2/xml/en/procedures/10/10-016-005-tr.html)
Procedure Title	Flywheel
Update Reason	Corrected torque sequence listed in third step of Install section. Updated flywheel mounting capscrews torque specifications.
Date	12-Dec-2022
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	10-019-443 (/qs3/pubsys2/xml/en/procedures/10/10-019-443.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Differential/Outlet Pressure Sensor
Update Reason	Added procedure to manual.
Date	14-Dec-2022
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	10-012-103 (/qs3/pubsys2/xml/en/procedures/10/10-012-103.html)
Procedure Title	Air Compressor Cylinder Head (Holset® SS and E-Type Models)
Update Reason	Added procedure to manual.

Date	14-Dec-2022
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	10-012-104 (/qs3/pubsys2/xml/en/procedures/10/10-012-104.html)
Procedure Title	Air Compressor Cylinder Head (Holset® QE Models)
Update Reason	Added procedure to manual.
Date	14-Dec-2022
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	10-012-106 (/qs3/pubsys2/xml/en/procedures/10/10-012-106.html)
Procedure Title	Air Compressor Cylinder Head (Holset® ST Models)
Update Reason	Added procedure to manual.
Date	21-Feb-2023
Section	Section 10 - Air Intake System - Group 10
Procedure Number	10-010-080-tr (/qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html)
Procedure Title	Air Intake Connection
Update Reason	Added torque in first Install stepblock.
Date	02-Mar-2023
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-056 (/qs3/pubsys2/xml/en/procedures/10/10-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Corrected exhaust gas pressure and intake manifold pressure specifications for a stationary (parked) regeneration.

Date	12-Jun-2023
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	101-019-449 (/qs3/pubsys2/xml/en/procedures/101/101-019-449.html)
Procedure Title	Aftertreatment Exhaust Gas Temperature Sensor
Update Reason	Added procedure to manual.
Date	04-Aug-2023
Section	Section 11 - Exhaust System - Group 11
Procedure Number	10-011-056 (/qs3/pubsys2/xml/en/procedures/10/10-011-056.html)
Procedure Title	Exhaust System Diagnostics
Update Reason	Updated specifications listed for a stationary (parked) regeneration.
Date	18-Aug-2023
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	10-019-444 (/qs3/pubsys2/xml/en/procedures/10/10-019-444.html)
Procedure Title	Aftertreatment Fuel Pressure Sensor
Update Reason	Added procedure to manual.
Date	18-Aug-2023
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	10-019-445 (/qs3/pubsys2/xml/en/procedures/10/10-019-445.html)
Procedure Title	Crankcase Pressure Sensor
Update Reason	Added procedure to manual.
Date	18-Aug-2023
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	10-019-365 (/qs3/pubsys2/xml/en/procedures/10/10-019-365.html)
Procedure Title	Crankshaft Position Sensor
Update Reason	Added procedure to manual.

Date	07-Sep-2023
Section	Section 14 - Engine Testing - Group 14
Procedure Number	10-014-027 (/qs3/pubsys2/xml/en/procedures/10/10-014-027.html)
Procedure Title	Aftertreatment Diesel Particulate Filter Regeneration Analyzer
Update Reason	Removed statement indicating the timing associated with the DPF regeneration analyzer test.
Date	24-Jun-2025
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	10-001-047-tr (/qs3/pubsys2/xml/en/procedures/10/10-001-047-tr.html)
Procedure Title	Piston Rings
Update Reason	Corrected typo found in procedure.
Date	26-Aug-2025
Section	Section 11 - Exhaust System - Group 11
Procedure Number	101-011-049-tr (/qs3/pubsys2/xml/en/procedures/101/101-011-049-tr.html)
Procedure Title	Aftertreatment Diesel Oxidation Catalyst
Update Reason	Added reference to Service Bulletin, California Aftertreatment Regulation Adherence, Bulletin 6642787.

Last Modified: 08-Dec-2025
